Q1) Let N denote the number of all 9 digits numbers if

a) the digit of each number are all from the set $\{5, 6, 7, 8, 9\}$ and

b) any digit that appears in the number repeats at least three times. Find the value of $\frac{N}{5}$.

Ans) 4201

Q2) A 4 digit number is called a doublet if any of its digit is the same as only one neighbour. For example 1221 is a doublet but 1222 is not. Find the number of such doublet.

Ans) 2268

Q3) Let N be the number of ordered pair (A, B) where A and B are subsets of $\{1,2,3,4,5\}$ such that neither $A \subseteq B$ nor $B \subseteq A$. Find N.

Ans) 570

Q4) Find the number of permutations $x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8$ of the integers $-3, -2, -1, 0, 1, 2, 3, 4$ that satisfy the chain of inequalities

$$x_1 x_2 \le x_2 x_3 \le x_3 x_4 \le x_4 x_5 \le x_5 x_6 \le x_6 x_7 \le x_7 x_8$$

Ans) 21

Q 5) In how many ways can we choose four 3-element subset of $\{1,2,3,4,5,6\}$ so that each pair of subsets share exactly one element.

Ans) 30